

Total No. of Questions : 10]

SEAT No. :

P4580

[Total No. of Pages : 2

[5669]-116

T.E. (Computer Engineering)

**DATABASE MANAGEMENT SYSTEMS APPLICATIONS
(2012 Pattern)**

Time : 2.½Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answers Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7, or Q.8, Q.9. or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Explain EER-Specialization with suitable example. [5]
b) Draw an ER diagram for Banking System. [5]

OR

- Q2)** a) Explain 3NF with Example. How transitive dependency is observed in database table? [5]
b) What is Data Isolation property of database? How it is achieved in DBMS [5]

- Q3)** a) Consider following structure for MongoDB collections and write a query for following requirements in MongoDB (Any 3)
Book(BookTitle, Author ISBN, Version, Year, Subject) [5]
i) Write a MongoDB query to create above collections.
ii) Find the information about all books of Author "Korth"
iii) Find all Authors who wrote database subject books
iv) Display Total no of books in 2019
b) How MongoDB handles the operation similar to group by operation in SQL? [5]

P.T.O.

OR

- Q4)** a) Differentiate Structured and Unstructured databases. [5]
b) What is Atomicity and Durability property of transaction? [5]

- Q5)** a) Explain Parallel Database Architectures in Detail. [8]
b) Explain Role of Transaction Manager and Transaction Coordinator in Distributed Transaction Management. [8]

OR

- Q6)** a) Explain different steps required for JAVA to MongoDB database connection using JDBC. [8]
b) There are different reasons of building distributed database systems, including sharing of data, autonomy and availability. Explain these advantages of distributed database systems in detail. [8]

- Q7)** a) Explain XQuery and FLWOR with Example. [7]
b) Explain JSON with example. [5]
c) Explain in brief different building blocks of HADOOP. [5]

OR

- Q8)** a) List & explain advantages of using XML Schema over XML DTD. [7]
b) Write a short note on: R programming. [5]
c) Write short note on Querying XML data. [5]

- Q9)** a) Explain Data Warehouse with its components? [5]
b) Explain Apriori Algorithm? Explain the term Support and Confidence. [7]
c) What is OLAP? Explain the OLAP operations in Details. [5]

OR

- Q10)** a) Write short note on Business Intelligence. [5]
b) How K means clustering works? How the Centroid of the cluster is evaluated? How Similarity between the data and Centroid is Evaluated? [7]
c) Differentiate Supervised and Unsupervised Learning with Example [5]

